

CHEMICAL BONDING AND MOLECULAR STRUCTURE

Chemical bonding is the force of attraction that holds atoms or ions together to form molecules or compounds. It arises due to the tendency of atoms to attain **stable electronic configuration**.

1 TYPES OF CHEMICAL BONDING

- Ionic Bond
- Covalent Bond
- Coordinate (Dative) Bond
- Metallic Bond
- Hydrogen Bonding & Van der Waals forces

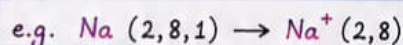


Why do atoms bond?

- To achieve stable noble gas configuration (octet / duet rule)
- To lower potential energy and gain stability.

Stable Configurations

Atoms react to achieve complete valence shell like noble gases.

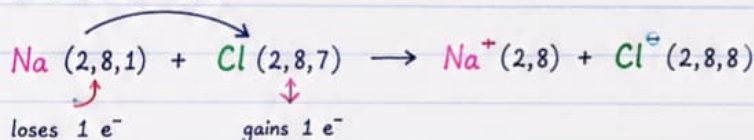


2 IONIC BOND

- Formed by the complete **transfer of electron(s)** from one atom (usually metal) to another atom (usually non-metal).
- Electrostatic attraction between **oppositely charged ions** holds them together.

Example

Formation of NaCl

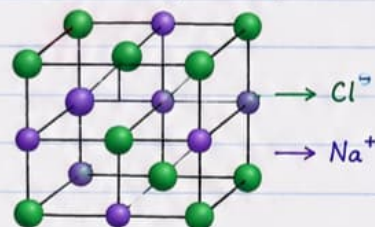


Na^+ and Cl^- attract each other $\rightarrow$ **NaCl** (ionic compound)

Properties of Ionic Compounds

- High melting & boiling points
- Hard but brittle
- Soluble in water
- Conduct electricity in molten state or in aqueous solution but not in solid state.

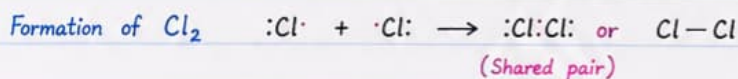
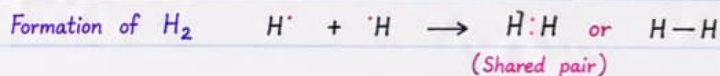
3D Structure of NaCl (Rock Salt)



3 COVALENT BOND

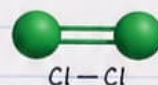
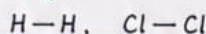
- Formed by the **sharing of one or more pairs of electrons** between two atoms (usually non-metals).
- Results in the formation of molecules.

Example

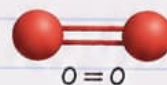
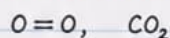


Types of Covalent Bonds

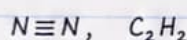
(i) Single Bond



(ii) Double Bond



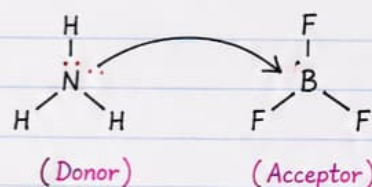
(iii) Triple Bond



4 COORDINATE (DATIVE) BOND

- Formed by the sharing of a pair of electrons in which both the electrons are provided by the same atom.
- The atom donating the pair is called **donor** and the atom accepting the pair is called **acceptor**.

Example: $\text{NH}_3 \rightarrow \text{BF}_3$



CHEMICAL BONDING AND MOLECULAR STRUCTURE

Key Idea

Atoms combine in different ways to attain stability and form structures with definite shape and properties.

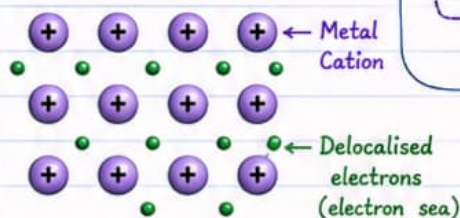
5 METALLIC BOND

- Formed between **metal atoms**.
- Valence electrons are delocalised over the whole lattice and form an "**electron sea**".
- Strong electrostatic attraction exists between the **positive metal ions** and delocalised electrons.

Properties of Metals

- ★ Good conductors of heat and electricity
- ★ Malleable and ductile
- ★ High melting and boiling points (generally)
- ★ Sonorous and lustrous

Metallic Lattice (e.g., Na)



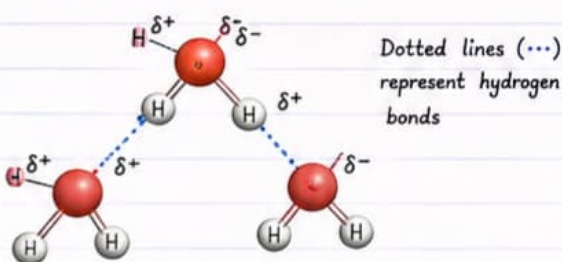
6 HYDROGEN BONDING ☆

- A **special** type of **dipole-dipole** attraction.
- Occurs when H atom covalently bonded to highly electronegative atoms (F, O or N) is attracted by another electronegative atom having a **lone pair**.

Importance of Hydrogen Bonding

- ★ High b.p. of water
- ★ Ice is less dense than water
- ★ High surface tension and viscosity
- ★ Anomalous behaviour of water

Example : Hydrogen bonding in water (H_2O)



Note

Hydrogen bond is weaker than covalent bond but stronger than van der Waals forces.

7 VSEPR THEORY (VALENCE SHELL ELECTRON PAIR REPULSION THEORY)

- The **shape** of a molecule is determined by the **repulsion** between electron pairs (bonding + lone pairs) in the valence shell.
- Electron pairs arrange themselves as far apart as possible to **minimise repulsion**.
- Shapes depend on the number of electron pairs around the central atom.

Remember *

Lone pairs exert greater repulsion than bonding pairs.

No. of Electron Pairs	Electron Pair Arrangement	Molecular Shape	Example	3D Structure	Bond Angle
2	Linear	Linear	$BeCl_2$		180°
3	Trigonal Planar	Trigonal planar	BF_3		120°
4	Tetrahedral	Tetrahedral	CH_4		109.5°
4	Tetrahedral	Trigonal pyramidal	NH_3		107°
4	Tetrahedral	Bent (V-shaped)	H_2O		104.5°
5	Trigonal bipyramidal	Trigonal bipyramidal	PCl_5		90° and 120°

Common Molecules and their Shapes

- CO_2 - Linear
- BF_3 - Trigonal planar
- CH_4 - Tetrahedral
- NH_3 - Trigonal pyramidal
- H_2O - Bent
- PCl_5 - Trigonal bipyramidal
- SF_6 - Octahedral (6 pairs)

Quick Tip ☆

More lone pairs → greater repulsion → smaller bond angle.

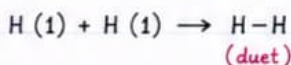
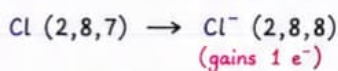
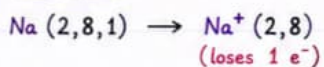
CHEMICAL BONDING AND MOLECULAR STRUCTURE

- Octet Rule
- Shapes of Molecules
- VSEPR Theory
- Hybridisation Basics

8 OCTET RULE

- Atoms tend to gain, lose or share electrons so that they attain the **stable noble gas configuration** (8 electrons in valence shell).
- **Exceptions**: H, He (duet stable), B (incomplete octet), 3rd period elements and beyond (expanded octet).

Examples



Why Octet?

Noble gases are least reactive because their valence shell is complete. Other atoms attain this stability.

9 SHAPES OF SOME SIMPLE MOLECULES (VSEPR)

Molecule	Electron Pair Arrangement	Molecular Shape	Example	3D Structure	Bond Angle
Linear	Linear	Linear	BeCl_2		180°
Trigonal planar	Trigonal planar	Trigonal planar	BF_3		120°
Tetrahedral	Tetrahedral	Tetrahedral	CH_4		109.5°
Trigonal pyramidal	Tetrahedral	Trigonal pyramidal	NH_3		107°
Bent	Tetrahedral	Bent (V-shaped)	H_2O		104.5°
Trigonal bipyramidal	Trigonal bipyramidal	Trigonal bipyramidal	PCl_5		90° (axial-equatorial 90° , equatorial-equatorial 120°)
Octahedral	Octahedral	Octahedral	SF_6		90°

Lone Pair – Lone Pair Repulsion >
Lone Pair – Bond Pair Repulsion >
Bond Pair – Bond Pair Repulsion

Tips to Predict Shape

1. Draw Lewis structure.
2. Count total electron domains (bonding + lone pairs).
3. Arrange domains to minimise repulsion.
4. Assign shape considering only atoms.

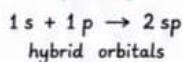
10 HYBRIDISATION

- Intermixing of atomic orbitals of nearly same energy on the same atom to form **equivalent hybrid orbitals**.
- Hybrid orbitals help in forming bonds with definite shapes.

Key Point

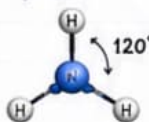
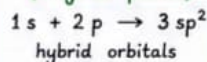
- More s-character $\rightarrow$ bond shorter & stronger.
- More p-character $\rightarrow$ bond longer & weaker.
- Hybrid orbitals are directional.

sp Hybridisation (Linear)



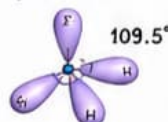
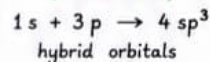
Example: BeCl_2

sp^2 Hybridisation (Trigonal planar)



Example: BF_3

sp^3 Hybridisation (Tetrahedral)

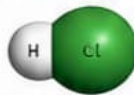


Example: CH_4

11 POLARITY OF BONDS AND MOLECULES

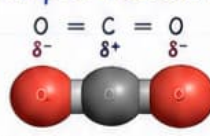
- Due to **difference in electronegativity** between bonded atoms.
- Polar molecule $\rightarrow$ has a **net dipole moment**.
- Non-polar molecule $\rightarrow$ net dipole moment = 0.

Polar Molecule (HCl)



Net dipole moment $\neq 0$

Non-polar Molecule (CO_2)



Net dipole moment = 0

Remember

Minimisation of electron pair repulsion leads to stable shapes of molecules.

NEET Tip

Focus on shapes, bond angles, hybridisation & polarity – frequently asked in NEET!

Common Exceptions to Octet Rule

- Incomplete Octet : BeCl_2 , BF_3
- Expanded Octet : PCl_5 , SF_6
- Odd Electron Molecules : NO , NO_2

CHEMICAL BONDING AND MOLECULAR STRUCTURE

NCERT Focus

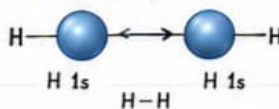
- Valence Bond Theory
- Molecular Orbital Theory
- Factors affecting Bond Length & Energy

12 VALENCE BOND THEORY (VBT)

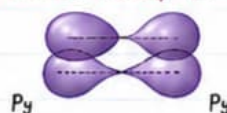
- Proposed by **Linus Pauling**.
- The bond is formed by the **overlap** of half filled atomic orbitals of two atoms.
- Overlap can be axial (end to end) or lateral (sidewise).
- Shapes and **magnetic behaviour** of molecules explained successfully.

Types of Overlap

(i) Axial Overlap (σ -bond)



(ii) Lateral Overlap (π -bond)



σ and π Bonds

σ (sigma) bond $\rightarrow$ Formed by head on overlap. Stronger bond.

π (pi) bond $\rightarrow$ Formed by sideways overlap. Weaker than σ -bond.

Examples (VBT)

(i) H_2 (Linear)

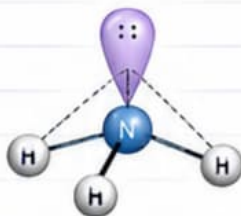
$1s - 1s$ axial overlap



Bond angle = 180°

(ii) NH_3 (Trigonal pyramidal)

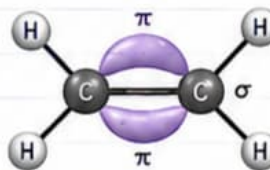
$sp^3 - 1s$ overlap



Bond angle $\approx 107^\circ$

(iii) C_2H_4 (Ethene)

$sp^2 - sp^2$ (σ) and $p-p$ (π) overlap



Bond angle $\approx 120^\circ$

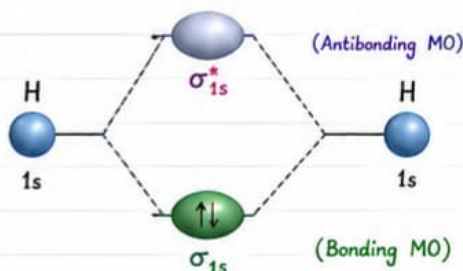
Merits of VBT

- ★ Explains shape of molecules.
- ★ Explains bond directionality.
- ★ Explains magnetic properties.
- ★ Explains unequal bond length in some molecules.

13 MOLECULAR ORBITAL THEORY (MOT)

- Proposed by **Friedrich Hund** and **R. S. Mulliken**.
- Atomic orbitals combine to form molecular orbitals which are **delocalised** over the whole molecule.
- Types of MOs :
 - (i) **Bonding MO** $\rightarrow$ lower in energy, stabilising
 - (ii) **Antibonding MO** $\rightarrow$ higher in energy, destabilising
- Bond order = $\frac{1}{2}$ (No. of e^- in bonding MO - No. of e^- in antibonding MO)

MO Formation in H_2 Molecule



Remember !

Higher bond order $\rightarrow$ stronger bond
Shorter bond length
More stability

14 FACTORS AFFECTING BOND LENGTH AND BOND ENERGY

(i) Bond Length

- ★ Decreases with increase in bond order.
- ★ Inversely proportional to bond energy.
- ★ Increases with increase in size of atoms.

Example (Bond order $\uparrow \rightarrow$ Bond length $\downarrow$)

Single	Double bond	Triple bond
$C-C$	$C=C$	$C \equiv C$
154 pm	134 pm	120 pm

($C-C > C=C > C \equiv C$ in bond length)

(ii) Bond Energy

- ★ Increases with increase in bond order.
- ★ Directly proportional to bond strength.
- ★ Decreases with increase in bond length.

Example (Bond order $\uparrow \rightarrow$ Bond energy $\uparrow$)

$C-C$	$C=C$	$C \equiv C$
348 kJ mol^{-1}	615 kJ mol^{-1}	839 kJ mol^{-1}

($C-C < C=C < C \equiv C$ in bond energy)

General Trend

Property	Trend
Bond order $\uparrow$	Bond length $\downarrow$ Bond energy $\uparrow$
Atomic size $\uparrow$	Bond length $\uparrow$ Bond energy $\downarrow$
Electronegativity difference $\uparrow$	Ionic character $\uparrow$ Covalent character $\downarrow$

Important Points

- More bond order $\rightarrow$ stronger, shorter and more stable bond.
- More ionic character $\rightarrow$ longer and weaker bond.
- Hybridisation determines shape and bond angles.
- Electronegativity difference decides the polarity of bonds.

Quick Revision

- ✓ Chemical bond $\rightarrow$ force of attraction
- ✓ Types $\rightarrow$ Ionic, Covalent, Coordinate, Metallic, Hydrogen, Van der Waals
- ✓ Octet rule $\rightarrow$ stability by noble gas configuration
- ✓ VSEPR $\rightarrow$ predicts shape based on repulsion
- ✓ VBT & MOT $\rightarrow$ explain bonding & shape of molecules
- ✓ Bond length & energy $\rightarrow$ depend on bond order, atomic size & EN difference